ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

NATURAL SCIENCES

PRACTICE EXAM 3

MARKS: 100 **TIME: 2 hours**

Covers all four Grade 8 strands: Life and Living (leaves, a variegated-leaf starch test, a pond ecosystem and energy pyramid, adaptations, useful and harmful micro-organisms), Matter and Materials (a heating curve, expansion, classifying substances, chemical formulae, a gas-producing reaction), Energy and Change (static charge, cells, meters in circuits, safety, reflection, refraction and colour) and Planet Earth and Beyond.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Show ALL calculations with formulae and units. A non-programmable calculator may be used.
4. Write in full sentences where explanations are asked for.
5. Diagrams are NOT necessarily drawn to scale.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which raw material for photosynthesis is absorbed by the roots?

A Oxygen B Water C Glucose D Carbon dioxide

1.2 The tiny openings in a leaf through which gases move in and out are called …

A stomata. B veins. C chloroplasts. D root hairs.

1.3 In a food chain, a green plant is a …

A consumer. B producer. C decomposer. D predator.

1.4 Athlete's foot is caused by a …

A virus. B bacterium. C fungus. D protist.

1.5 The change from a liquid to a gas is called …

A melting. B freezing. C evaporation. D condensation.

1.6 In which of these are the water particles moving fastest?

A Ice B Cold water C Steam D Frost

1.7 How many different elements are there in sodium chloride (NaCl)?

A 1 B 2 C 3 D 4

1.8 Two objects with negative charges will …

A attract each other. B repel each other. C have no effect on each other. D become neutral.

1.9 In a parallel circuit, if one bulb blows, the other bulbs …

A go out. B stay on.
C become much brighter. D start to flicker.

1.10 The force that keeps the planets in their orbits around the Sun is …

A friction. B magnetism. C gravity. D static electricity.

(10)

QUESTION 2: TRUE OR FALSE

Write TRUE or FALSE next to the question number. If the statement is FALSE, write down the correct word(s) to replace the underlined word(s).

2.1 Most photosynthesis takes place in the roots of a plant. (2)

2.2 Viruses can be killed by antibiotics. (2)

2.3 The particles of a gas are far apart. (2)

2.4 An ammeter is used to measure electric current. (2)

2.5 The Sun is a planet. (2)

(10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3: LIFE AND LIVING

3.1 Give TWO ways in which the shape of a leaf helps it to photosynthesise. (2)

3.2 A variegated leaf has green parts and white parts. After it was tested with iodine, only the green parts turned blue-black. Explain this result. (2)

3.3 Study the information about a pond ecosystem: Algae are eaten by tadpoles and snails. Tadpoles are eaten by small fish. Snails are eaten by ducks. Small fish are eaten by kingfishers.

3.3.1 Name the producer in this ecosystem. (1)

3.3.2 Name ONE secondary consumer. (1)

3.3.3 Draw a labelled energy pyramid for the food chain: algae → tadpoles → small fish → kingfisher. (3)

3.3.4 Explain how ONE human activity could harm this pond ecosystem. (2)

3.4 Give TWO adaptations that help a cactus to survive in a desert. (2)

3.5 Describe how bacteria are used to make yoghurt from milk. (3)

3.6 Explain why washing your hands with soap helps to prevent illness. (2)

3.7 Give TWO differences between viruses and bacteria. (2)

[20]

QUESTION 4: MATTER AND MATERIALS

4.1 Crushed ice at −10 °C was heated in a beaker and the temperature was recorded every 2 minutes.

Time (min)	0	2	4	6	8	10	12	14
Temperature (°C)	−10	0	0	40	80	100	100	100

4.1.1 At what temperature does ice melt? (1)

4.1.2 Explain why the temperature stays the same between 2 and 4 minutes, even though heat is still being added. (2)

4.1.3 What is the boiling point of water according to the results? (1)

4.1.4 In what state is the water at 6 minutes? (1)

4.2 Explain why small gaps are left between the sections of a concrete bridge. (2)

4.3 Classify each substance as an element, a compound or a mixture: (a) salt water; (b) copper; (c) sugar ($C_{12}H_{22}O_{11}$); (d) air. (4)

4.4 The formula for glucose is $C_6H_{12}O_6$. (a) How many different elements does it contain? (b) How many atoms are there in one molecule of glucose? (2)

4.5 Classify each element as a metal or a non-metal: (a) sodium; (b) sulfur; (c) magnesium; (d) oxygen. (2)

4.6 When vinegar is added to baking soda in an open beaker, the mixture fizzes.

4.6.1 Name the gas that is produced. (1)

4.6.2 Describe a test for this gas and give the result. (2)

4.6.3 The beaker and its contents weigh less after the reaction. Explain why. (1)

4.7 Is dissolving sugar in tea a physical change or a chemical change? (1)

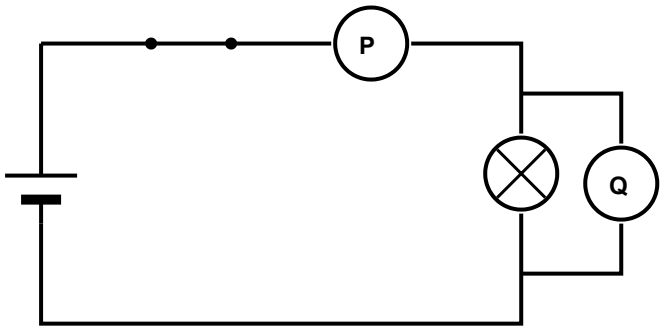
[20]

QUESTION 5: ENERGY AND CHANGE

5.1 A plastic ruler that has been rubbed with a woollen cloth can pick up small pieces of paper. Explain why. (3)

5.2 What happens to the brightness of a bulb when a second cell is added in series to the circuit? Explain. (2)

5.3 Study the circuit diagram.



P and Q are meters.

5.3.1 Meter P is connected in series. Name meter P and what it measures. (2)

5.3.2 Meter Q is connected in parallel across the bulb. Name meter Q and what it measures. (2)

5.4 Give TWO dangers of plugging too many appliances into one wall socket. (2)

5.5 (a) A ray of light strikes a flat mirror at an angle of 40° to the surface of the mirror. Calculate the angle of incidence and the angle of reflection. (b) Why can we see the Moon even though it does not produce its own light? (3)

5.6 A swimming pool looks shallower than it really is. Name and explain this effect. (2)

5.7 Which colour of light is bent the most when white light passes through a prism? (1)

5.8 Name the THREE primary colours of light. (3)

[20]

QUESTION 6: PLANET EARTH AND BEYOND

6.1 Why is it very difficult to draw an accurate scale model of the solar system on a sheet of paper? (2)

6.2 Name the FOUR gas giants. (2)

6.3 Give TWO reasons why the Earth can support life. (2)

6.4 Why do stars seem to twinkle when we look at them from Earth? (2)

6.5 Arrange these stages in the life of a star like our Sun in the correct order: red giant, white dwarf, nebula, main-sequence star (3)

6.6 The Andromeda Galaxy is about 2,5 million light-years away. What does this tell us about the light from Andromeda that we see tonight? (2)

6.7 Give TWO reasons why South Africa is a good place for astronomy. (2)

6.8 Give TWO reasons why astronauts must wear space suits when they work outside a spacecraft. (2)

6.9 Explain the difference between a meteor and a meteorite. (2)

6.10 Name the very large, ancient meteorite impact structure found in the Free State. (1)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF QUESTION PAPER